Introduction to Computer Science

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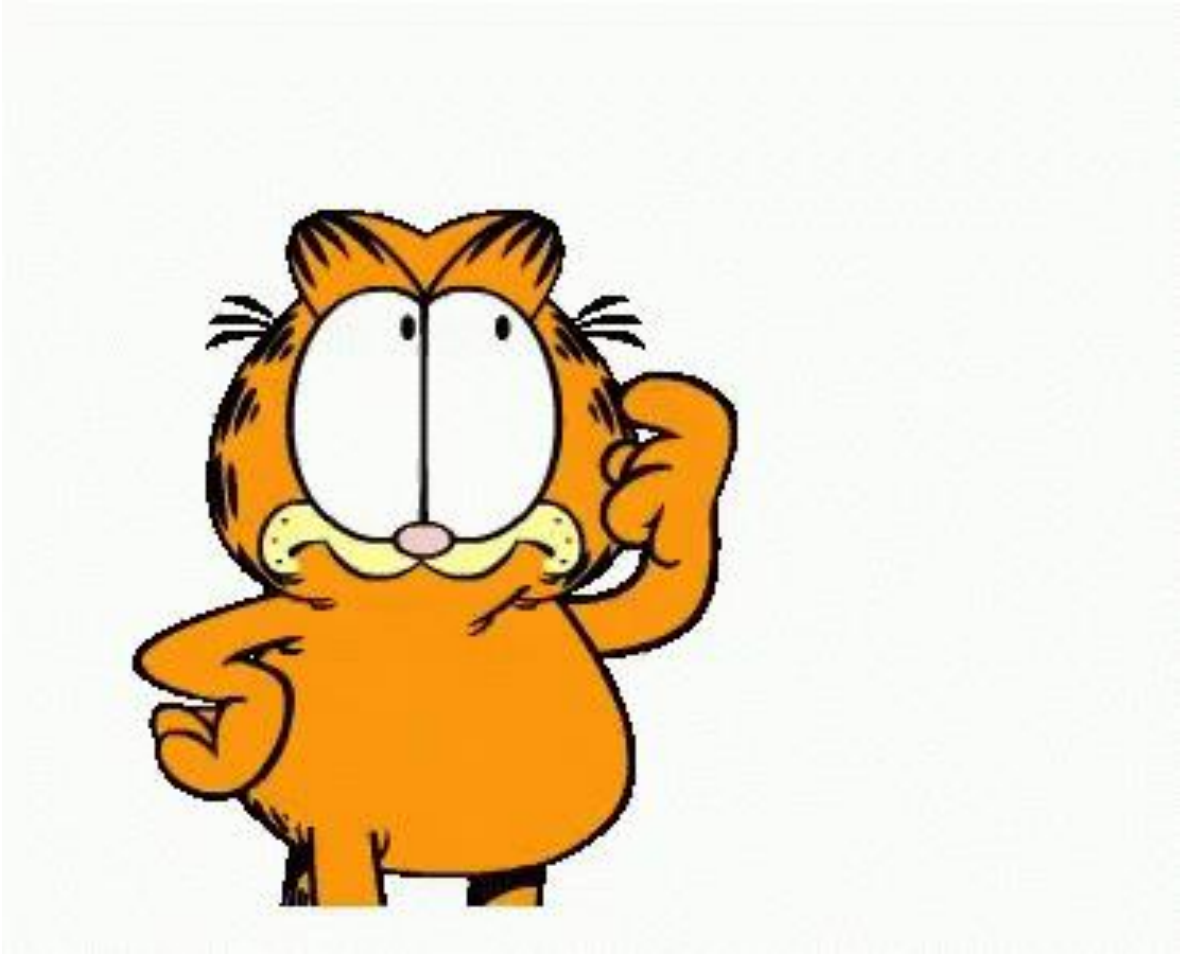
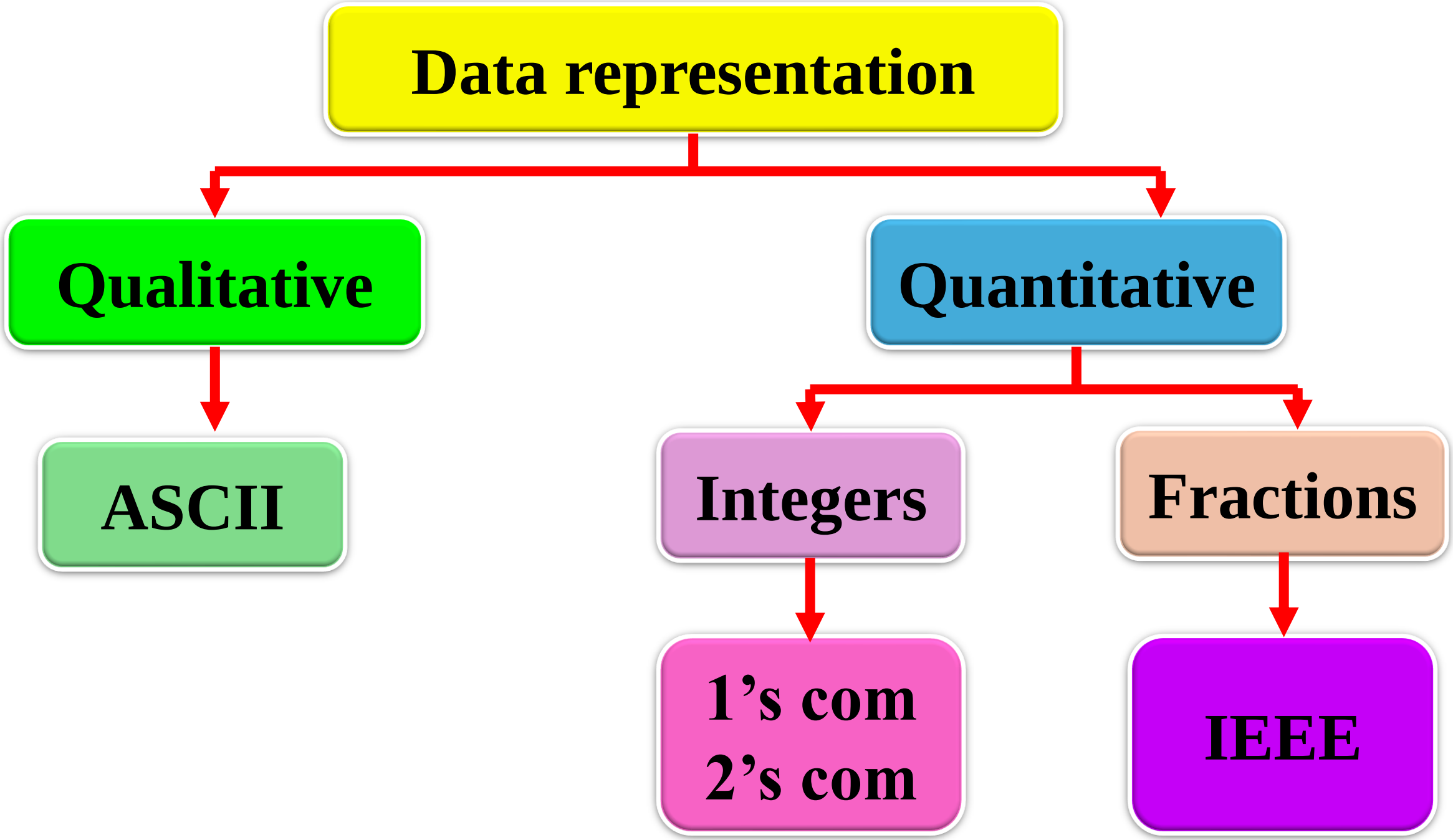
Data Representation in CS



What is Data Representation?

- A computer **represents a piece of data with a specific number of bits**, which can be a number, a character, an image, a sound, a video, etc.
- A computer's **internal data representation method**
- The methods **used to represent information in computers**
- The computer system can store a variety of data types.


Data representation
Qualitative
Quantitative
ASCII
Integers
Fractions
1’s com
2’s com
IEEE

Qualitative Data Representation

Qualitative Data Representation

- **ASCII – American Standard Code for Information Interchange**
- **Character encoding standard used to represent text in computers**
- **Includes printable characters such as letters of English alphabet (upper and lower case), numbers, and punctuation marks.**
- **Each ASCII character is assigned a unique number between 0 and 127.**

ASCII

Symbol	Decimal	Binary
A	65	01000001
B	66	01000010
C	67	01000011
D	68	01000100
E	69	01000101
F	70	01000110
G	71	01000111
H	72	01001000
I	73	01001001
J	74	01001010
K	75	01001011
L	76	01001100
M	77	01001101
N	78	01001110
O	79	01001111
P	80	01010000
Q	81	01010001
R	82	01010010
S	83	01010011
T	84	01010100
U	85	01010101
V	86	01010110
W	87	01010111
X	88	01011000
Y	89	01011001
Z	90	01011010

Symbol	Decimal	Binary
a	97	01100001
b	98	01100010
c	99	01100011
d	100	01100100
e	101	01100101
f	102	01100110
g	103	01100111
h	104	01101000
i	105	01101001
j	106	01101010
k	107	01101011
l	108	01101100
m	109	01101101
n	110	01101110
o	111	01101111
p	112	01110000
q	113	01110001
r	114	01110010
s	115	01110011
t	116	01110100
u	117	01110101
v	118	01110110
w	119	01110111
x	120	01111000
y	121	01111001
z	122	01111010

ASCII – Reading alphabet characters

Example : COMPUTER

C	O	M	P	U	T	E	R
67	79	77	80	85	84	69	82

Activity

- **Write the ASCII code for following.**

1. **Computer Science**
2. **I love NIBM**
3. **Data Representation**

ASCII – Hexa-Decimal Code

		000	001	010	011	100	101	110	111
		0	1	2	3	4	5	6	7
0000	0	NULL	DLE		0	@	P	`	p
0001	1	SOH	DC1	!	1	A	Q	a	q
0010	2	STX	DC2	"	2	B	R	b	r
0011	3	ETX	DC3	#	3	C	S	c	s
0100	4	EDT	DC4	$	4	D	T	d	t
0101	5	ENQ	NAK	%	5	E	U	e	u
0110	6	ACK	SYN	&	6	F	V	f	v
0111	7	BEL	ETB	'	7	G	W	g	w
1000	8	BS	CAN	(	8	H	X	h	x
1001	9	HT	EM	)	9	I	Y	i	y
1010	A	LF	SUB	*	:	J	Z	j	z
1011	B	VT	ESC	+	;	K	[	k	{
1100	C	FF	FS	,	<	L	\	l	|
1101	D	CR	GS	-	=	M	]	m	}
1110	E	SO	RS	.	>	N	^	n	~
1111	F	SI	US	/	?	O	_	o	DEL

Reading ASCII Hexa-Decimal Code

1.Read the column first and the row second of the relevant character.

Activity

- **Identify the message written using ASCII Hexadecimal code**

1. **49 20 4C 6F 76 65 20 49 54**
2. **48 65 79 21 49 20 66 6F 75 6E 64 20 69 74 2E**
3. **44 61 6E 67 65 72 6F 75 73 21**
4. **49 20 4C 6F 76 65 20 49 43 53**

Activity

- **Identify the message written using ASCII Hexadecimal code**

1. **I Love my life.**
2. **Happy Halloween!**
3. **Good Morning!**
4. **See you…**

Quantitative Data Representation

Fractional

IEEE 754

Converting floating point number into Binary

- To convert the fractional part to binary, **multiply fractional part with 2 and take the one bit which appears before the decimal point.**

- **Follow the same procedure with after the decimal point (.) part until it becomes 1.0.**

Example

4.25

4 . 25

Integral part

Decimal point

Decimal part

4.25 – 100.01_2

	25	X 2
0	50	X 2
1	0	

Decimal part

2	4	
2	2	0
	1	0

Integral part

Example 2

5.375 - 101.011_2

2	5	
2	2	1
	1	0

	375 X 2
0	750 X 2
1	500 X 2
1	0

Activity

Covert below floating-point number to binary.

1. 25.75
2. 7.125
3. 151.375

Converting Binary to Decimal

Example : 100.101

1	0	0	.	1	0	1
2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}
				1/2	1/4	1/8

$100.101_2 - 4.625_{10}$

Activity

Covert below binary to decimal.

1. 1101.01

2.110101.11

3.10111.101